Diagnosis-Guided Compressed Latent Replay for Continual Document Understanding

Supplementary Material

A. Setup details

DIL unified schema. A single four-entity label space {HEADER, KEY, VALUE, OTHER} (9 BIO tags) is held fixed over FUNSD→SROIE→CORD. FUNSD: HEADER→HEADER, QUESTION→KEY, ANSWER→VALUE. SROIE: COMPANY→KEY; ADDRESS/DATE/TOTAL→VALUE. CORD: menu/sub_total/total→VALUE; void_menu/sub→OTHER. KEY originates only from FUNSD and SROIE, so the schema is VALUE-dominant; we report per-class F1 for every DIL result and treat a high aggregate as retention only if the sparse classes are retained. The reverse order is used as a robustness check.

CIL-CORD sessions. CORD's 30 fine-grained classes in five sessions of six, ordered easy-to-hard by superclass: (0) core menu fields; (1) extended menu + sub; (2) sub_total fields; (3) total / payment methods; (4) residual total + void/auxiliary. Every receipt containing at least one in-session entity is kept, with out-of-session tokens relabelled to "O", giving per-session training pools of 800/382/549/723/780 receipts. Class imbalance across sessions is deliberately left unmodified. Joint uses the full-label pool (each receipt once with all its labels) rather than the concatenation of the masked per-session copies, which would feed conflicting labels for the same document.

Mixed. Six sessions: CORD-menu → FUNSD → CORD-menu-extended → SROIE → CORD-sub_total → FUNSD-revisit; boundaries alternate new-class and new-domain transitions and the final session is an explicit retention check.

Datasets. FUNSD (`nielsr/funsd-layoutlmv3`; 149/50), CORD (`naver-clova-ix/cord-v2`; 800/100/100), SROIE (`mp-02/sroie`; 626/347). Dataset-provided OCR and boxes throughout; images 224×224; sequence length 512.

Hyperparameters. AdamW, lr 5×10^{-5} constant, weight decay 0.01, clipping 1.0, batch size 8 on the primary grid (1–2 with gradient checkpointing on the 6 GB secondary-backbone machine; replay batch 2, or 1 for DER++ on BROS, whose HF implementation lacks checkpointing). EWC λ=10^3, Fisher over 200 documents; LwF T=2, α=1; ER/DER++ buffer 200 documents, DER++ α=β=0.5. Early stopping on task val F1, patience 2, δ=0.1, cap 100 epochs (never binding; tasks converge in 4–8). Seeds 42/7/123. Primary-grid document runs used bf16 autocast on a 24 GB GPU; the secondary-backbone cells ran in fp32 or fp16 on a 6 GB GPU. Image CIL (CIFAR-100, ImageNet-R; ViT-B/16, 10 sessions): every reported cell, including the frozen-backbone and replay baselines, was trained on a 6 GB GPU at batch size 16 in fp32 with gradient checkpointing, 100-epoch cap with the same early stopping, seeds 42/7/123; the new CODA-Prompt, CL-LoRA and LCA image cells use this recipe unchanged.

Table 1. Prompt/LoRA/C-Flat++ baselines, DIL, AA and BWT, mean±std over seeds 42/7/123. Naive floor 41.3 (LayoutLMv3), ER 87.9.

	LayoutLMv3		BERT	
Method	AA	BWT	AA	BWT
L2P	30.4 ± 0.4	-11.5 ± 0.1	34.4 ± 0.7	-38.5 ± 1.0
DualPrompt	29.9 ± 0.6	-11.1 ± 0.2	34.0 ± 0.2	-40.2 ± 1.1
CODA-Prompt	32.5 ± 1.2	-10.5 ± 0.2	35.2 ± 0.7	-40.7 ± 1.3
O-LoRA	40.8 ± 0.3	-56.8 ± 1.0	37.4 ± 0.3	-53.4 ± 0.9
CL-LoRA	41.1 ± 0.3	-58.8 ± 2.1	38.6 ± 0.4	-31.6 ± 1.9
ER + C-Flat++	88.4 ± 0.1	-1.8 ± 0.6	78.3 ± 0.4	-2.2 ± 0.1

Compute. Every run fits a single GPU; the full generality grid is roughly 200 GPU-hours on a 24 GB card.

B. Additional results

Prompt-, LoRA- and flatness-based baselines. Tab. 1 reports the frozen-backbone prompt family (L2P, DualPrompt, CODA-Prompt), the orthogonal/continual LoRA family and ER with C-Flat++ sharpness-aware optimisation on DIL, three seeds, LayoutLMv3 and BERT. The prompt family forgets little but never acquires the task (AA ≈ 30–35, below the naive floor of 41.3): a frozen document encoder with a drifting shared head cannot be routed back into place. LoRA adapters acquire but forget as much as naive training does, because the adapter does not protect the head. Only the replay-based variant reaches the replay ceiling, and its sharpness term adds nothing over plain ER (88.4 vs. 87.9 on LayoutLMv3), consistent with Sec. 5 of the main paper: the remedy ordering is set by whether the head is re-grounded, not by the optimiser.

Per-backbone forgetting. The full BWT and AF tables for every backbone×scenario×method cell of the generality grid are generated from the released run directories (`results/table_backbone_{BWT,AF}.tex`)

and will ship with the code; Table 1 of the main paper already carries the AA view.

Budget-adaptive allocation is a frontier, not a gain. Allocating a fixed compressed-replay budget unevenly across tasks (documents $[18, 5, 2]$, ranks $[64, 128, 64]$ vs. a uniform 10 documents at rank 64; 5 epochs, k=4) gave $+4.0$ AA on seed 42 while passing a pre-registered per-domain guard. On seeds 7 and 123 the same comparison gave -8.0 and $+5.7$: $+0.6 \pm 7.5$ over three seeds, a null. The seed-42 result was a point on a Pareto frontier (old-domain retention traded against the final domain at $\approx 9{:}1$), not a shift of the frontier, and we report it as such.

Null cells among the fixed baselines (CPFD, IS3, CODA-Prompt). Three of the twelve reference rows do not converge on the class-incremental document scenarios, and we report those cells as nulls rather than as scores. CPFD and IS3 were ported onto the single growing token-classification head used by every other method here (the reference implementations keep per-task classifiers). On CIL-CORD, CPFD learns the first session (90–95 F1) and then reads 0.00 on every session from the first class-adding boundary (AA 2.7 and 0.0 on seeds 42 and 123); on Mixed it reads 6.0, 0.0 and 0.1. Two single-knob ablations on seed 42 reproduce the collapse exactly: flooring the per-sample CE weight at 0.5 (session-1 F1 0.00) and switching the distillation term off entirely (session-1 F1 2.1), so neither the adaptive CE factor nor the attention/logit distillation is the cause; what remains is the pseudo-labelling and O-token masking path, which removes the training signal on the new session's entities. IS3 learns each new session (diagonal $93/29/32/66/10$ on seed 123) but never predicts a previously learned class again—every off-diagonal entry is exactly 0.00—giving AA 4.4 ± 1.7 on CIL-CORD; on Mixed, whose last session adds no classes, it recovers to 30.1 ± 1.0 and that cell is reported. CODA-Prompt reads AA 0.0 on all three CIL-CORD seeds with its default pool of 10 components; enlarging the pool to 50 (10 per session) lets some sessions be learned (diagonal $79.5/40.6/0.0/18.9/36.8$) but the final row is $[0, 0, 0, 0, 36.8]$, AA 7.4, with one session never learned, so the cell stays a null. These are properties of the methods' O-tag and static-pool assumptions under five class-adding sessions, not of the sessions themselves: the same sessions are learned by naive training (AA 19.0) and by the joint upper bound (33.3).